

Endogenous Firm Scope via Organizational Complexity

Sebastian Dyrda

University of Toronto

Guangbin Hong

Michigan State University

Joseph B. Steinberg

University of Toronto & NBER

Society for Economic Dynamics Annual Meeting

Athens, Greece

July 2, 2026

Scalable Capability, Bounded Scope

- The **rise of intangibles** makes capability easy to scale: build the blueprint once, deploy it everywhere at near-zero cost. (Argente, Moreira, Oberfield & Venkateswaran 2025; Crouzet, Eberly, Eisfeldt & Papanikolaou 2024)
- Yet realized scope is **bounded** and **related**: firms add nearby markets, mostly through R&D and acquisitions. (Hoberg & Phillips 2025)
- Even Amazon (healthcare) and Google (hardware) **stall in distant markets**. If capability replicates so freely, why doesn't a handful of firms span the whole economy?

What **bounds firm scope?**

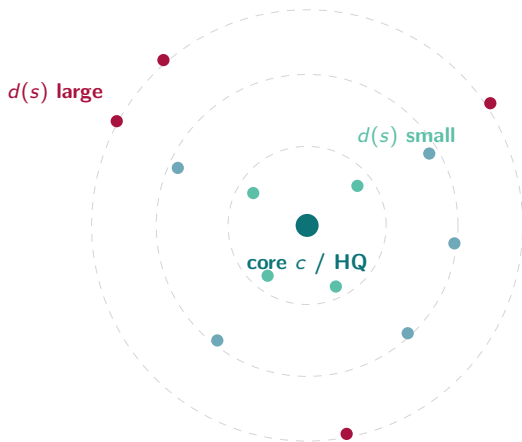
This Paper: Scope Bounded by Complexity, Not Count

The binding constraint is not the number of markets a firm serves but how **different** they are. We capture this as **organizational complexity**: the cumulative heterogeneity across a firm's footprint.

- Distant markets are organizationally heterogeneous; adapting the non-rival blueprint there consumes scarce headquarters attention, so capability does not translate into **scope**.
- Expansion is **non-separable**: adding a market raises the cost of serving every other. (unlike count-based theories: span of control, Lucas 1978; entry costs, Melitz 2003; knowledge hierarchies, Garicano 2000, Garicano & Rossi-Hansberg 2006)
- Against the winner-take-all intuition, the model **predicts** local dominance **need not aggregate**: scalable capability need not concentrate the economy.

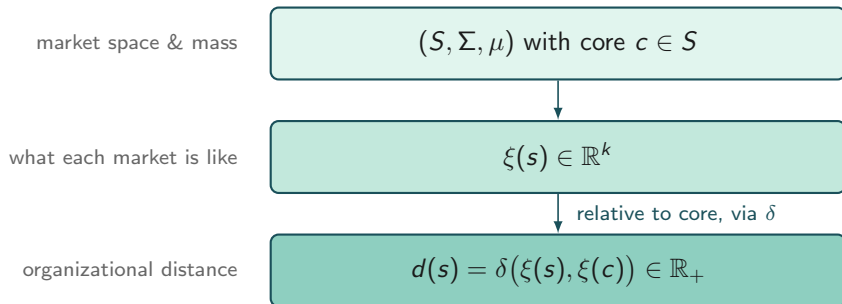
Environment

A Firm in One Picture



- One **non-rival blueprint**, created at the core, deployed across a footprint of **heterogeneous markets**.
- Each market sits at organizational distance $d(s)$ from the core.
- Deploying to distant markets must be adapted; adaptation consumes **rival headquarters attention**.

Markets and Heterogeneity



μ records the **mass of markets**, ξ records **what they are**, and $d(s)$ is a market's **organizational distance** from the core. Each market also carries exogenous capacity $\bar{X}(s) > 0$. [► details](#)

Organizational Complexity

Integrating organizational distance over the footprint gives a single firm-level state:

Definition. For any measurable footprint $M \subseteq S$,

$$C(M) \equiv \int_M d(s) \mu(ds).$$

Complexity is driven by **cumulative heterogeneity**, **not the number of markets**.

► [details](#)

The Blueprint

A firm operates a centralized blueprint $z \geq 0$, created at headquarters:

- **Non-rival**: one blueprint z serves all markets at once.
- **Centralized**: created and maintained at headquarters.
- **Adapted**: each deployment must be customized locally.

Creating quality z requires headquarters attention

$$z = \Gamma(h_z) \quad \Longleftrightarrow \quad \chi(z) \equiv \Gamma^{-1}(z) \text{ units of attention,}$$

with Γ continuous, strictly increasing, $\Gamma(0) = 0$.

Deployment and Returns

Effective deployment of the blueprint in market s :

$$z_s = \theta(\underset{+}{a}(s), \underset{-}{d}(s)) z, \quad a(s) = A(\underset{+}{h}(s), \underset{+}{\ell}(s))$$

θ : portability A : adaptation

Effective deployment and local input generate the market-level payoff:

$$R(\underset{+}{z}_s, \underset{+}{x}(s))$$

$h(s)$: headquarters attention $\ell(s)$: local effort $x(s)$: local production input

► details

The Central Friction: Rival Headquarters Attention

Headquarters attention is **rival** within the firm: split between blueprint creation and adaptation. Serving a more heterogeneous footprint makes it more expensive everywhere.

Cost of headquarters attention:

$$q(C(M)) \left(\underbrace{\chi(z)}_{\text{creation}} + \underbrace{\int_M h(s) \mu(ds)}_{\text{adaptation}} \right)$$

Organizational congestion function:

- $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$, $q(0) = 1$, $q' > 0$, $q(C) \rightarrow \infty$.
- q is an ordinary congestion wedge. The economic content is the **state it congests on**: cumulative heterogeneity $C(M)$ (not size!).

The Firm's Problem

The firm chooses blueprint quality z , footprint M , and allocations $\{x(s), \ell(s), h(s)\}$:

$$\max_{z, M, \{x, \ell, h\}} \int_M R(z_s, x(s)) \mu(ds) - q(C(M)) \left(\chi(z) + \int_M h(s) \mu(ds) \right)$$

$$\text{s.t. } x(s) + \ell(s) \leq \bar{X}(s) \quad \forall s \in M$$

$$\text{where } z_s = \theta(a(s), d(s)) z$$

$$M = \{s : x(s) + \ell(s) + h(s) > 0\}$$

Baseline Results

Endogenous Scope Limit

Fix capability z . What bounds the firm's reach?

Proposition (bounded complexity). At any (minimal-support) optimum,

$$C(M^*(z)) = \int_{M^*(z)} d(s) \mu(ds) < \infty.$$

- Bounded **cumulative heterogeneity** but **not** the mass $\mu(M^*)$, which may be infinite.
- A finite mass of **materially distinct** markets ($d \geq \tau$); an unbounded mass of near-core ones.

Expansion Is Non-Separable

Let $\Pi_s(z; C)$ be the value of operating market s at complexity C .

Proposition (organizational congestion). $\Pi_s(z; C)$ is weakly decreasing in C ; hence adding markets Δ weakly lowers every incumbent:

$$\Pi_s(z; C(M \cup \Delta)) \leq \Pi_s(z; C(M)) \quad \forall s \in M.$$

- A market's value depends on the **whole footprint**, through C . Thus selection is non-separable.
- Separable theories (Lucas–Garicano–Melitz) **cannot generate this**.

Related Expansion: A Cutoff in Heterogeneity

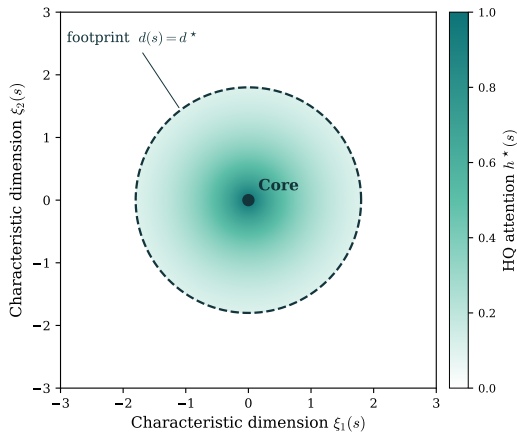
Which markets does the firm serve?

Proposition (cutoff selection). Under a heterogeneity-ordering condition, there is a threshold d^* such that the firm serves exactly the markets within it:

$$M^*(z) = \{ s : d(s) \leq d^* \}.$$

- Firms serve markets **near the core** and exclude those beyond the threshold d^* (related expansion, Hoberg & Phillips 2025).
- **No fixed costs or decreasing returns**: selection comes from **organizational scarcity**.
- Threshold rests on a **heterogeneity-ordering condition**, e.g. uniform local capacity.
▶ details

Core-Periphery Organization



$h^*(s)$ concentrates at the core, a gradient from [organizational scarcity](#).

Endogenous Blueprint: Returns to Scale

Now let the firm choose blueprint quality z . The blueprint is **non-rival**. How does scope shape the incentive to invest in it?

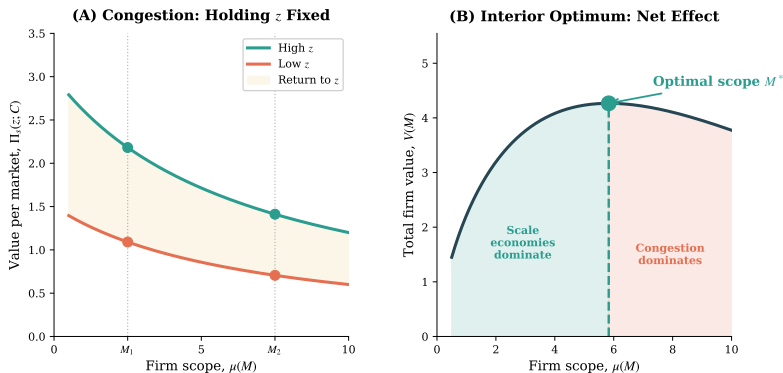
Proposition (returns to blueprint scale). For any $z_2 \geq z_1$ and $M_1 \subseteq M_2$,

$$\int_{M_2} [\Pi_s(z_2; C) - \Pi_s(z_1; C)] \mu(ds) \geq \int_{M_1} [\Pi_s(z_2; C) - \Pi_s(z_1; C)] \mu(ds).$$

- The return to a better blueprint **rises with the served set**. Every market deploys the same blueprint, amplifying the gain from improving it.
- Scope and blueprint quality are **complements**: a wider footprint $\Rightarrow$ stronger incentive to raise z .

Two Forces Shape Firm Scope

Expanding the footprint $M \rightarrow M' = M \cup \Delta$ pits two forces against each other:



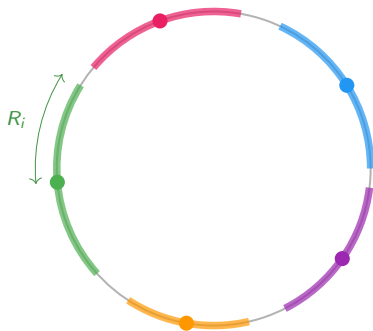
$$\Delta \Pi = \underbrace{[\Pi_s(z^*(M'); C(M')) - \Pi_s(z^*(M); C(M'))]}_{\text{blueprint adjustment } (\geq 0)} + \underbrace{[\Pi_s(z^*(M); C(M')) - \Pi_s(z^*(M); C(M))]}_{\text{organizational congestion } (\leq 0)}$$

Local Dominance & Concentration

What the single-firm scope limit implies across many firms.

A Multi-Firm Environment

Embed many firms on a shared space: the torus $\mathbb{T}_L$, a circle of circumference L .



- Core c_i (**dot**) and footprint $M_i = \{s : \delta(s, c_i) \leq R_i\}$ (**arc**).
- **Scope radius** $R_i = \mathcal{R}(z_i)$ (optimal reach), bounded by organizational complexity.
- Each firm **dominates locally** near its core.

Does this local dominance add up to **aggregate concentration**?

► details

Local and Aggregate Concentration: One Object

Firm i 's local activity share is $\omega_i(s)$. The **overlap kernel**:

$$\mathcal{O}(x) = \frac{1}{L} \int_{\mathbb{T}_L} \sum_i \omega_i(s) \omega_i(s+x) ds$$

Proposition (overlap-kernel identity).

$$\mathcal{O}(0) = \overline{\text{HHI}}_{\text{loc}} \equiv \frac{1}{L} \int_{\mathbb{T}_L} \sum_i \omega_i(s)^2 ds,$$

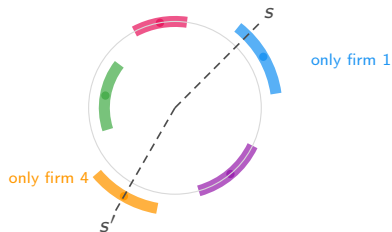
$$\frac{1}{L} \int_{\mathbb{T}_L} \mathcal{O}(x) dx = \text{HHI}_{\text{agg}} \equiv \sum_i \bar{\omega}_i^2.$$

where $\bar{\omega}_i = \frac{1}{L} \int_{\mathbb{T}_L} \omega_i$ is the average share, and HHI is the Herfindahl–Hirschman index.

- How fast $\mathcal{O}$ decays governs the gap between **local** and **aggregate** concentration.

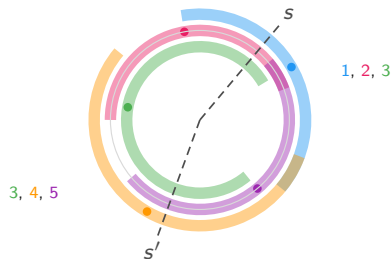
Scope Compression Lowers Overlap

Organizational complexity



compressed arcs, little overlap

Without scope compression



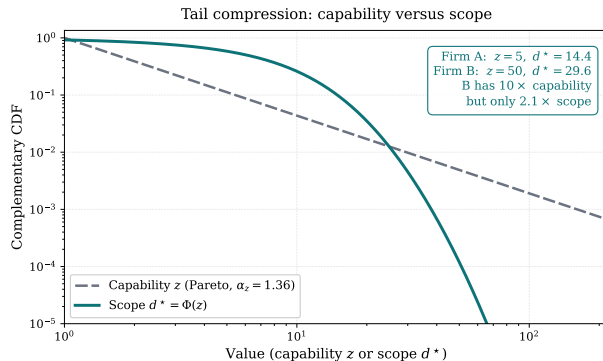
wide arcs, same firm spans s, s'

- Compressed scope $\Rightarrow \mathcal{O}$ decays fast: each firm dominates near its core, with little overlap elsewhere.
- **Local dominance does not aggregate:** high local HHI, low aggregate HHI. Scalable capability need not concentrate the economy.

Quantitative Illustration

Tail Compression and Missing Giants

Calibrate the model to Hoberg–Phillips scope: (mean 7.6, median 6.0, sd 5.8) with blueprints z distributed according to Pareto. [► details](#)

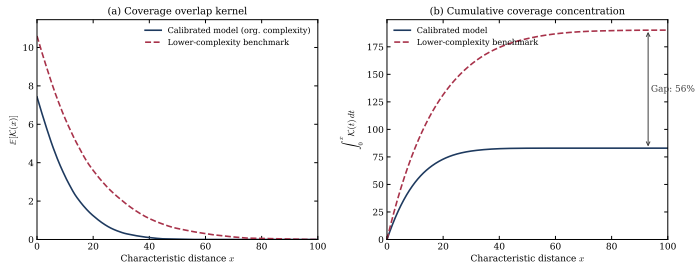


Pareto capability ($\alpha_z = 1.36$) $\Rightarrow$ a thinner-tailed scope distribution: $10 \times$ capability buys only $\sim 2.1 \times$ scope: the “missing giants.”

Local Dominance, Bounded Aggregate Concentration

Embed the calibrated radii on a torus and compare with a **lower-complexity benchmark**.

► details



$\mathcal{K}(x)$ is the **coverage kernel**: same-firm presence at distance x , the extensive-margin counterpart of $\mathcal{O}$.

Complexity makes coverage decay faster: integrated overlap is **56% lower**.

Local dominance does not aggregate: no firm spans enough distant markets.

Key Takeaways

1. Scope is bounded by **organizational complexity**: the cumulative heterogeneity of a firm's markets, not their count.
2. Expansion is **non-separable**: scope is the net of blueprint scalability and organizational congestion.
3. Capability does not buy scope one-for-one: adapting to heterogeneous markets compresses the scope tail, the **missing giants**.
4. **Local dominance need not aggregate**: a firm dominates near its core while aggregate concentration stays bounded.

Appendix: Technical Assumptions

Assumption: Non-atomic Market Space

◀ back

Assumption (Non-atomic market space). (S, Σ, μ) is non-atomic: for every $E \in \Sigma$ with $\mu(E) > 0$ and every $\varepsilon \in (0, \mu(E))$, there exists $E' \subseteq E$, $E' \in \Sigma$, with $\mu(E') = \varepsilon$.

Rules out indivisible markets of positive mass: the firm can expand or contract its footprint by arbitrarily small amounts. The market space behaves like a continuum (Lebesgue on an interval), the natural idealization for scope spanning many small, related activities.

Assumption: Positive Heterogeneity

◀ back

Assumption (Positive heterogeneity). $d(s) > 0$ for μ -a.e. $s \in S$.

Rules out a positive-mass cluster of markets identical to the core. The normalization $d(c) = 0$ is consistent with this since the core is a null reference point. The firm may still operate across vast collections of nearly core-aligned markets with arbitrarily small $d(s)$.

Example (organizational distance). A natural form is a weighted norm,

$$d(s) = \left(\sum_{j=1}^k \omega_j |\xi_j(s) - \xi_j(c)|^p \right)^{1/p},$$

but no result requires $d(\cdot)$ to satisfy the axioms of a norm.

Assumptions: Deployment and Effectiveness

◀ back

Assumption (Adaptation and portability). $A : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ continuous, weakly increasing in each argument. $\theta : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow [0, 1]$ continuous, weakly increasing in a , weakly decreasing in d , with $\lim_{a \rightarrow \infty} \theta(a, d) = 1$ for all $d \geq 0$.

Assumption (No free deployment). For any s with $d(s) > 0$,

$$\theta(A(0, \ell), d(s)) = 0 \quad \text{for all } \ell \geq 0.$$

Assumption (Blueprint effectiveness). $R(z_s, x)$ weakly increasing in each argument.

An Inada-type portability technology: zero deployment without headquarters attention, full deployment in the limit of unbounded adaptation.

Assumptions: Congestion and Integrability

◀ back

Assumption (Organizational congestion). $q : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ with $q(0) = 1$; continuously differentiable and strictly increasing; $\lim_{C \rightarrow \infty} q(C) = \infty$.

Broader, more heterogeneous scope raises coordination burdens, increasing the marginal cost of headquarters attention anywhere in the firm, including blueprint creation $\chi(z)$.

Assumption (Conditional integrability). For fixed blueprint quality z , gross payoffs are dominated by some $\bar{R}_z \in L^1(S, \mu)$, ruling out unbounded payoffs at fixed technology.

Appendix: The Heterogeneity-Ordering Condition

◀ back

Assumption (Heterogeneity ordering). For any fixed (z, C) , the market value $\Pi_s(z; C)$ is weakly decreasing in $d(s)$:

$$d(s_1) \leq d(s_2) \implies \Pi_{s_1}(z; C) \geq \Pi_{s_2}(z; C).$$

More heterogeneous markets are weakly less valuable to serve; this monotonicity is what turns selection into a threshold in d .

Sufficient primitive condition. Uniform local capacity $\bar{X}(s) = \bar{X}$: the inner feasible set is common across markets, effective deployment $\theta(A(h, \ell), d(s))$ z falls in d , so the objective $R(z_s, x) - q(C)h$ is pointwise larger at smaller d ; taking suprema delivers the ordering.

Sufficient, not necessary: the ordering also holds for non-uniform $\bar{X}$ co-varying suitably with d . It restricts neither returns to scale nor within-market technology.

Proposition: Core–Periphery Allocation

◀ back

Single-crossing (decreasing differences). With

$V(h, d; z) \equiv \sup_{x+\ell \leq \bar{X}} R(\theta(A(h, \ell), d) z, x)$, for $h_1 \geq h_0$ and $d_1 \leq d_2$,

$$V(h_1, d_1; z) - V(h_0, d_1; z) \geq V(h_1, d_2; z) - V(h_0, d_2; z).$$

Proposition (core–periphery allocation). Under single-crossing, constant local capacity $\bar{X}(s) = \bar{X}$, and single-valued $h^*(s)$: for any $s_1, s_2 \in M$,

$$d(s_1) \leq d(s_2) \implies h^*(s_1) \geq h^*(s_2).$$

Headquarters attention is weakly decreasing in heterogeneity: the incremental payoff from attention is higher in low- d (core) markets.

Appendix: The Torus Environment

[◀ back](#)

Characteristic space. Torus $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$ of circumference L , with torus metric $\delta(s, s')$.

Firms. Firm i : core c_i , blueprint z_i . By the cutoff result (Prop 2.3) the optimal footprint is the arc

$$M_i = \{s \in \mathbb{T}_L : \delta(s, c_i) \leq R_i\}, \quad R_i = \mathcal{R}(z_i) = \Phi(z_i)/2,$$

with scope radius induced by the scope function Φ of the distribution section.

No-double-overlap: $R_i \leq L/4$: arcs short enough that two shifted copies overlap only on the nearest side. **Cores** $\{c_i\}$ are exogenous (no entry, exit, or placement modeled).

Appendix: Local Shares and the Two Kernels

◀ back

Local share (zero-activity convention): $\omega_i(s) = y_i(s)/Y(s)$ if $Y(s) > 0$, else 0; $\sum_i \omega_i(s) \in \{0, 1\}$.

$$\text{HHI}(s) = \sum_i \omega_i(s)^2, \quad \overline{\text{HHI}}_{\text{loc}} = \frac{1}{L} \int \text{HHI}, \quad \text{HHI}_{\text{agg}} = \sum_i \bar{\omega}_i^2, \quad \bar{\omega}_i = \frac{1}{L} \int \omega_i.$$

Two kernels. Share-weighted $\mathcal{O}$ (genuine HHI, Prop 4.1) vs. coverage $\mathcal{K}$ (extensive-margin co-presence):

Coverage formula (Prop 4.2). For interval footprints with $R_i \leq L/4$,

$$\mathcal{K}(x) = \frac{1}{L} \sum_i (2R_i - x)_+, \quad \mathbb{E}[\mathcal{K}(x)] = \frac{N}{L} \mathbb{E}[(2\mathcal{R}(Z) - x)_+].$$

The calibration's $\mathbb{E}[\mathcal{K}(x)]$ (and the 56% gap) is the extensive-margin object; the genuine-HHI statement runs through the share-weighted $\mathcal{O}$.

Appendix: Scope Compression Reduces Overlap

◀ back

Proposition (scope compression reduces coverage overlap). If $\mathcal{R}(z) \leq \tilde{\mathcal{R}}(z) \leq L/4$ (F_z -a.e., strict on a positive-measure set), then

$$\mathbb{E}[\tilde{\mathcal{K}}(x)] \geq \mathbb{E}[\mathcal{K}(x)] \quad \forall x, \quad \mathbb{E}\left[\frac{1}{L} \int_0^{L/2} \tilde{\mathcal{K}}\right] > \mathbb{E}\left[\frac{1}{L} \int_0^{L/2} \mathcal{K}\right].$$

Appendix: Calibration

◀ back

	Target (HP)	Model
<i>Targeted moments</i>		
Mean scope	7.55	7.36
Median scope	6.00	6.09
Std. dev. of scope	5.80	5.77
<i>Calibrated parameters</i>		
Pareto tail index α_z	—	1.36
Portability decay λ	—	18.11
Congestion weight κ	—	0.005
Congestion elasticity φ	—	1.96
<i>Non-targeted</i>		
P90 / P99 scope	—	15.0 / 26.1
Skewness	—	1.47

$H(d) = \exp(d/\lambda)(1 + \kappa d^2/2)^\varphi$, Pareto z . **Caveat:** 3 targets, 4 parameters: a disciplined illustration of magnitudes, not separately identified primitives.

Appendix: The Lower-Complexity Benchmark

◀ back

Construction. Both economies share the calibrated capability distribution and the same torus ($L = 300$, uniform cores). They differ only in the **congestion elasticity** φ of the scope technology:

$$H(d) = \exp(d/\lambda) (1 + \kappa d^2/2)^\varphi, \quad R = \mathcal{R}(z) = \Phi(z)/2, \quad \Phi = H^{-1}.$$

- Calibrated economy: $\varphi = 1.96$, so organizational complexity bites.
- **Lower-complexity benchmark:** $\varphi_{\text{low}} = 0.5$, a weaker congestion penalty, so scope radii are larger and less compressed.

A **lower-complexity** benchmark, not frictionless linear scaling: scope is still capability-bounded, only the complexity penalty is softened. The 56% is the gap in integrated coverage $\frac{1}{L} \int_0^{L/2} \mathcal{K}$ between the two economies.